

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“JNANA SANGAMA”, BELAGAVI-590018



VTU Internship Report ON

“Android App Development using Gen AI”

submitted in partial fulfillment of the requirement for the award of the degree of

BACHELOR OF ENGINEERING in ELECTRONICS AND COMMUNICATION ENGINEERING

SUBMITTED BY:

**SHREYA RAJENDRA DEVEKAR
(1SP22EC051)**

Under the Guidance of

Internal Guide: Mr. B. R. Narendra
Internship Coordinator,
Assistant Professor,
Dept. of ECE, SEACET

External Guide: Mr. Tirumal Mutalikdesai
Program Coordinator,
MindMatrix



**DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
S.E.A COLLEGE OF ENGINEERING AND TECHNOLOGY
BENGALURU-560049**

S.E.A. COLLEGE OF ENGINEERING AND TECHNOLOGY

BENGALURU-560049

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING



CERTIFICATE

I, Certified that Internship Report entitled "**Android App Development using Gen AI**" is a Bonafide work carried out by **Shreya Rajendra Devekar (ISP22EC051)** carried out by are the bonafide student of **S.E.A. College of Engineering and Technology**, in partial fulfillment for the award of **BACHELOR OF ENGINEERING in ELECTRONICS AND COMMUNICATION ENGINEERING**, **Visvesvaraya Technological University**, Belagavi during the year 2025-26. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the departmental library. The Internship report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.

Guide

Mr. B. R. Narendra
Asst. Professor
SEACET, Bengaluru

HOD

Dr. Pradeep Kumar
Head of Department
SEACET, Bengaluru

Principal

Dr. B Venkata Narayana
Principal,
SEACET, Bengaluru

Name of the Examiners

External Viva

Signature with date:

1.

2.

DECLARATION

I, Shreya Rajendra Devekar (ISP22EC051), student of 8th semester BE in Electronics and Communication Engineering, of S.E.A. College of Engineering and Technology, Bengaluru. hereby declare that the project work entitled “**Android App Development using Gen AI**” submitted to the **Visvesvaraya Technological University, Belgavi** during the academic year 2025-26, is a record of an original work done by us under the guidance of **Mr. B. R. Narendra**, Department of Electronics & Communication Engineering, S.E.A. College of Engineering and Technology, Bengaluru. This internship report is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Engineering in Electronics & Communication Engineering. The results embodied in this report have not been submitted to any other University or Institute for the award of any degree.

Place:

Bangalore

Date:

NAME

USN

SIGNATURE:

Shreya Rajendra Devekar

(ISP22EC051)

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Cordially,

Shreya Rajendra Devekar (1SP22EC051)

ABSTRACT

This report details a comprehensive 15-week project-based internship completed at MindMatrix focusing on 'Android App Development using Gen AI'. The objective of this programme was to bridge the academia-industry gap by transitioning engineering students from a theoretical mindset to a professional 'System Builder' mentality. The technical phase involved creating high-performance Android applications using Kotlin, Jetpack Compose, Room Database, and Retrofit for network integration. A major highlight of the internship was integrating the Gemini API via Google AI Studio for rapid prototyping and AI-grounded responses within the application. The programme deeply emphasized industrial project management, utilizing the DMAIC framework, Git version control, and the 'Critical Path' approach for performance optimization using the Google Antigravity tool. Beyond technical skills, the internship cultivated the 'Campus to Corporate' (C2C) framework, instilling professional ethics, structured communication using the PREP method, and problem-solving through 5 Whys and Fishbone analysis. The internship effectively built the core competencies required for real-world software engineering and prepared the intern for a sustainable professional career.

Keywords: Android Development, Kotlin, Jetpack Compose, Generative AI, Gemini API, Room Database, Retrofit, DMAIC, Git, MindMatrix.

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CHAPTER 1

ABOUT THE ORGANIZATION

CHAPTER - 1



ABOUT THE ORGANIZATION

1.1 Introduction to the Organization

Mind Matrix is an Information Technology and educational training organization that provides practical technical learning and industry-oriented internship programs for engineering students. The organization mainly focuses on Android application development, Artificial Intelligence integration, software development training, and project-based learning methodologies. The internship programs conducted by Mind Matrix are designed to help students gain practical exposure to modern technologies and real-time software development environments.

The organization emphasizes hands-on implementation instead of only theoretical learning. Students are encouraged to develop applications, maintain GitHub repositories, create project documentation, and present their work professionally. This approach helps students understand how software development workflows are implemented in industry environments.

1.2 Vision of the Organization

The vision of Mind Matrix is to create technically strong and industry-ready professionals with practical implementation skills and modern software development knowledge. The organization aims to bridge the gap between academic education and industry requirements through project-oriented technical learning. The organization also focuses on promoting innovation, problem-solving ability, and AI-assisted development practices among students so that they can adapt to modern technological advancements.

1.3 Mission of the Organization

The mission of Mind Matrix is to provide students with quality technical education, real-time project exposure, and practical software development experience. The organization aims to improve students' programming skills, development workflows, debugging ability, project management understanding, and technical confidence through structured mentorship and practical activities. The organization also encourages collaborative learning, technical communication, and continuous skill development through workshops, coding sessions, and project implementation tasks.

1.4 Domains and Services

Mind Matrix mainly works in the following technical domains:

- Android Application Development
- Artificial Intelligence and AI-assisted Development
- UI/UX Design and Frontend Development
- Educational Technology Solutions
- Software Development Training
- Project Mentorship and Guidance

The organization also provides exposure to modern development tools and platforms such as Android Studio, Kotlin, GitHub, Google AI Studio, ChatGPT, and Antigravity.

1.5 Training Methodology

The internship training methodology followed a structured learning approach where concepts were introduced gradually from basic to advanced levels. Initially, students were trained in Android Studio setup, Kotlin basics, and UI design fundamentals. Later, students implemented mini-projects and advanced applications to improve practical coding skills.

The training process included:

- Daily technical sessions
- Practical coding implementation
- Project-based learning
- Mentor interaction and support
- GitHub portfolio management
- AI-assisted development workflows
- Debugging and testing activities

This structured approach helped students improve both theoretical understanding and practical software development skills.

1.6 Importance of the Internship Program

The internship program provided practical exposure to modern software development technologies and workflows. Students learned how applications are planned, designed, developed, tested, and maintained using professional development methodologies.

The internship also improved awareness regarding:

Real-time project workflows

Industry-oriented technical practices

Project planning and requirement analysis

Team communication and collaboration

AI-assisted productivity improvement

CHAPTER 2

OBJECTIVES OF INTERNSHIP

CHAPTER-2

OBJECTIVES OF INTERNSHIP

2.1 Primary Objective

The primary objective of this internship was to gain practical exposure to Android application development using Kotlin and Android Studio. The internship focused on improving practical software development skills through continuous implementation and project-based learning. The internship also aimed to improve understanding of modern Android development technologies and UI design methodologies.

2.2 Technical Learning Objectives

The internship mainly focused on improving technical knowledge in the following areas:

Kotlin Programming Language

Android Studio Development Environment

Jetpack Compose UI Design

Android Application Lifecycle

Emulator and Device Testing

GitHub Repository Management

Debugging and Error Resolution

Students were encouraged to implement projects independently and improve their practical coding ability through regular practice.

2.3 Android Development Objective

Another important objective was to understand Android application structure, navigation, layouts, and user interaction handling. Students learned how to create activities, manage layouts, implement navigation, and design interactive user interfaces.

The internship also introduced students to:

Material Design principles

Application architecture basics

UI responsiveness

User-friendly application design

This helped students understand frontend application development workflows.

2.4 AI-Assisted Development Objective

The internship aimed to improve understanding of AI-assisted development workflows using tools such as ChatGPT and Antigravity. Students learned how AI tools can help in:

Code generation

UI prototyping

Debugging support

Productivity improvement

Error analysis

Learning assistance

This objective helped students understand the practical role of Artificial Intelligence in modern software development environments.

2.5 Project Development Objective

The internship also focused on improving project implementation and management skills. Students learned how to:

Understand project requirements

Prepare project documentation

Design application UI screens

Develop application logic

Test applications

Maintain GitHub repositories

The internship also introduced Agile methodology concepts, sprint planning, requirement engineering, and project workflow management practices.

2.6 Professional Development Objective

Apart from technical skills, the internship also aimed to improve professional skills such as:

Communication skills

Problem-solving ability

Technical presentation skills

Documentation practices

Logical thinking

Time management

These skills are essential for software development careers and professional environments.

CHAPTER 3

LEARNING EXPERIENCES

CHAPTER-3

LEARNING EXPERIENCES

3.1 Introduction to Learning Experience

The internship provided extensive practical learning experiences through technical sessions, coding implementation activities, project reviews, AI-assisted workflows, and debugging exercises. The learning process followed a gradual progression from basic Android concepts to advanced application development methodologies.

3.2 Android Studio and Kotlin Learning

Initially, students learned how to install and configure Android Studio and create Android projects using Kotlin programming language. Basic Android concepts such as:

Activities

Layouts

UI components

Navigation

Event handling

Application lifecycle

were introduced through practical coding implementation.

Students also learned how to use Android emulator and real Android devices for application testing.

3.3 UI/UX Design Learning

The internship emphasized modern UI/UX design principles using Jetpack Compose. Students learned:

Layout design techniques

Text styling and alignment

Color combinations

Responsive UI implementation

User-friendly interface design

Google AI Studio was also used to generate frontend UI screens through prompts and design suggestions.

Students were encouraged to observe real-world applications and implement professional UI practices in their projects.

3.4 Project Implementation Learning

Several applications were developed during the internship to improve practical implementation skills. These projects include:

Hello Android Application

Dice Roller Application

Bit Calculator Application

Namaz Timing Application

Product Analyzer Application

Student Productivity Application

Each project introduced new technical concepts and improved understanding of Android application workflows.

3.5 Agile and Project Management Learning

The internship also included sessions on:

Agile Methodology

Scrum Framework

Sprint Planning

User Stories

Use Cases

Kanban Workflow Management

Requirements Engineering

Students learned how software projects are planned and managed using structured development methodologies. The mentors explained how Product Requirement Documents (PRD), use cases, and project scopes are prepared before application development.

3.6 AI-Assisted Workflow Learning

AI tools such as ChatGPT and Antigravity were actively used during the internship for:

Code suggestions

Error resolution

UI generation

Debugging support

Workflow improvement

Students learned how AI can improve productivity and simplify software development workflows.

3.7 GitHub and Portfolio Development Learning

Students were instructed to upload project progress regularly to GitHub and maintain professional repositories. GitHub Pages were used for portfolio development and project showcasing.

This improved understanding regarding:

Version control

Repository management

Project backup

Portfolio creation

Professional project presentation.

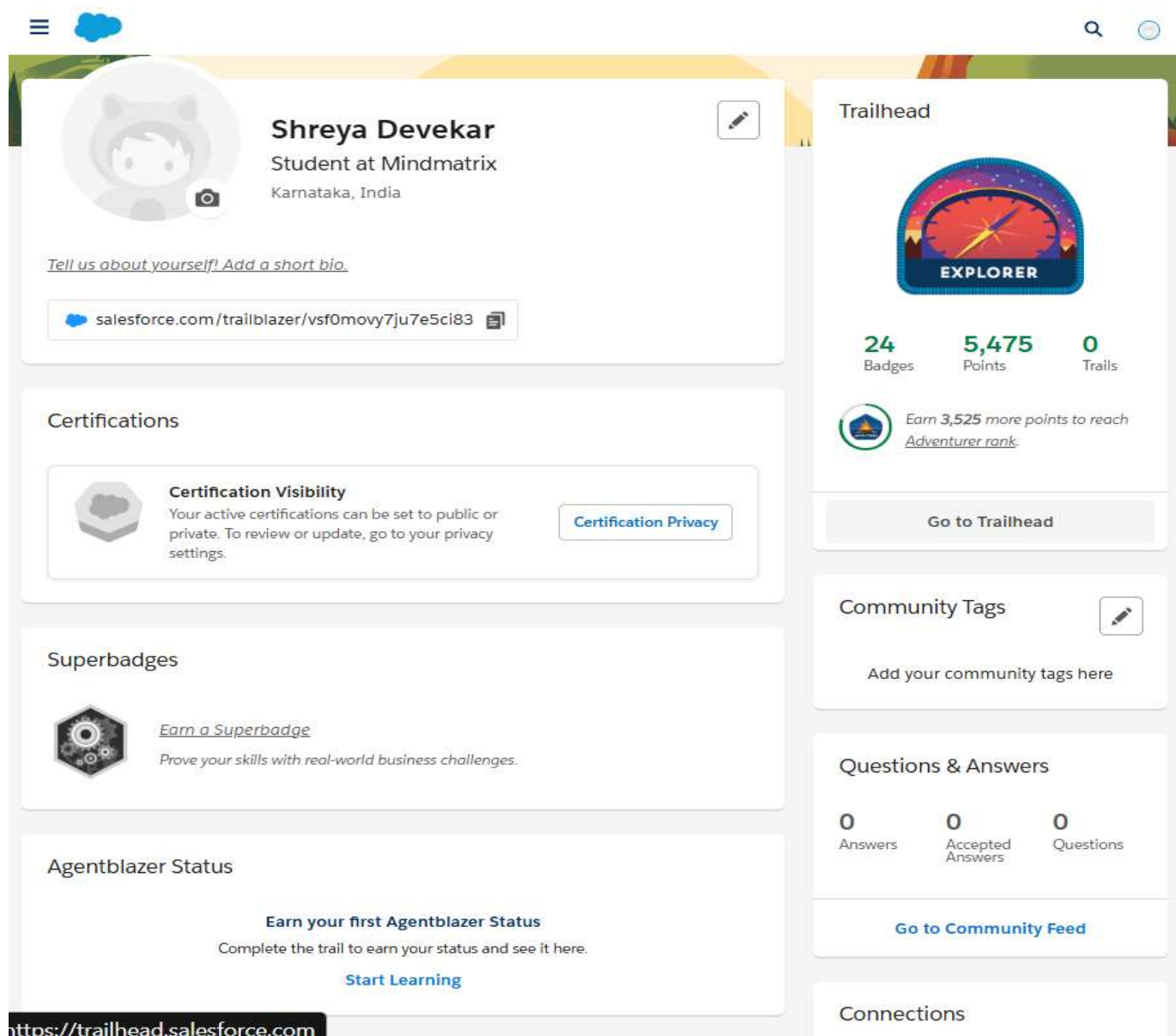


Figure 3.7.1: Salesforce Trailblazer Profile

CHAPTER-4

LEARNING OUTCOMES

CHAPTER- 4

LEARNING OUTCOMES

During this internship, I gained practical knowledge and hands-on experience in Android application development using Kotlin and Android Studio. I learned how to design and develop user interfaces using Jetpack Compose and understood the basics of Android application structure, activity lifecycle, layouts, navigation, and UI components. Through multiple mini-projects such as Dice Roller, Bit Calculator, Namaz Timing App, and Student Productivity App, I improved my understanding of real-world application development and step-by-step implementation of features.

I also gained experience in using development tools such as Android Studio, Android Emulator, and GitHub for project management and version control. I learned how to test applications on both emulator and real Android devices, which helped me understand application compatibility and debugging techniques. During development, I faced issues related to Gradle builds, emulator configuration, and runtime errors, which improved my problem-solving and troubleshooting skills.

Another important learning outcome of this internship was understanding AI-assisted development workflows. I used tools like ChatGPT and Antigravity for idea generation, debugging support, UI visualization, and understanding coding concepts more effectively. This helped me improve productivity and learn modern development approaches used in the software industry.

Overall, this internship enhanced my technical knowledge, logical thinking, debugging ability, and confidence in Android application development. It also gave me exposure to industry-oriented workflows, collaborative learning, project planning, and continuous skill improvement through practical implementation.

CHAPTER-5

PROJECT DESCRIPTION

CHAPTER - 5**PROJECT DESCRIPTION****5.1 Introduction**

Construction work in rural and small-scale sectors is mainly handled by local builders and masons, commonly called mistris. These workers are responsible for estimating construction materials, managing laborers, tracking site activities, and maintaining the overall workflow of construction projects. In most cases, these activities are performed manually using notebooks, rough calculations, and verbal communication. Because of this traditional approach, mistakes frequently occur in material estimation and labor management, which leads to financial loss, project delays, and unnecessary wastage of resources.

In many construction projects, mistris estimate the number of bricks, cement bags, and sand quantity based on experience rather than accurate calculations. This often results in either overestimation or underestimation of materials. Overestimation causes wastage and increased construction cost, while underestimation delays the project and affects work efficiency. Along with this, labor attendance and payment details are also managed manually, making it difficult to maintain accurate records and calculate worker balances properly. To solve these problems, the proposed system “Namma-Mistri” is developed as an Android-based construction assistant application. The application is designed to help local builders and rural mistris manage construction activities in a more organized and professional way. The application provides features such as material estimation, labor diary management, site progress tracking, and standard material rate maintenance. By using this system, mistris can perform calculations accurately and maintain proper records digitally using a mobile device.

The project mainly focuses on simplifying daily construction management activities through technology. The application acts as a practical and easy-to-use digital tool for local builders who may not have access to expensive construction management software. It combines Android application development with construction-related calculation logic to create a useful real-world solution.

5.2 Motivation

In rural and small-town construction environments, most builders still depend completely on manual methods for planning and management. Material estimation is usually based on assumptions and personal experience, which can easily result in calculation errors. Labor attendance and payment records are commonly written in notebooks, and site progress is not maintained systematically. These issues create confusion, increase

workload, and reduce overall project efficiency. Many local builders are not familiar with advanced digital tools, and professional construction management software is often too expensive or complicated for small-scale usage. Therefore, there is a strong need for a simple, affordable, and mobile-based application that can help local mistris manage their work more effectively.

The motivation behind developing Namma-Mistri is to provide a digital construction assistant that is practical, user-friendly, and suitable for real construction-site conditions. The project aims to support local builders by helping them estimate materials accurately, maintain labor records properly, and improve overall construction planning through technology.

5.3 Problem Statement

Local builders and mistris face many challenges in managing construction work because most calculations and records are maintained manually. Construction material estimation is often inaccurate due to rough assumptions and lack of proper planning tools. This causes material wastage, unnecessary expenses, and project delays.

Labor management is another major issue faced at construction sites. Worker attendance, wages, advance payments, and balance calculations are usually maintained manually, which creates confusion and increases the possibility of financial errors. Site progress documentation is also not maintained properly, making it difficult to monitor construction development and communicate updates to clients or house owners.

Hence, there is a requirement for a simple Android application that can assist local builders in estimating materials accurately, maintaining labor records digitally, tracking work progress, and organizing construction activities efficiently.

5.4 Objectives of the Project

The main objective of the project is to develop an Android application that acts as a digital construction assistant for local builders and Mistri. The system aims to simplify construction-site management activities and improve accuracy in material estimation.

The application is designed to help users calculate the quantity of construction materials such as bricks, cement, and sand based on wall or room dimensions. It also aims to provide a proper labor management system where attendance, wages, advance payments, and balance due can be maintained digitally. Another important objective is to improve construction transparency by allowing users to store and monitor site progress photographs.

The project also focuses on creating a simple and user-friendly interface that can be easily operated by local builders without technical difficulty. Overall, the objective is to reduce manual effort, minimize errors, save time, and improve construction management efficiency using Android technology.

5.5 Android App Development Using Kotlin

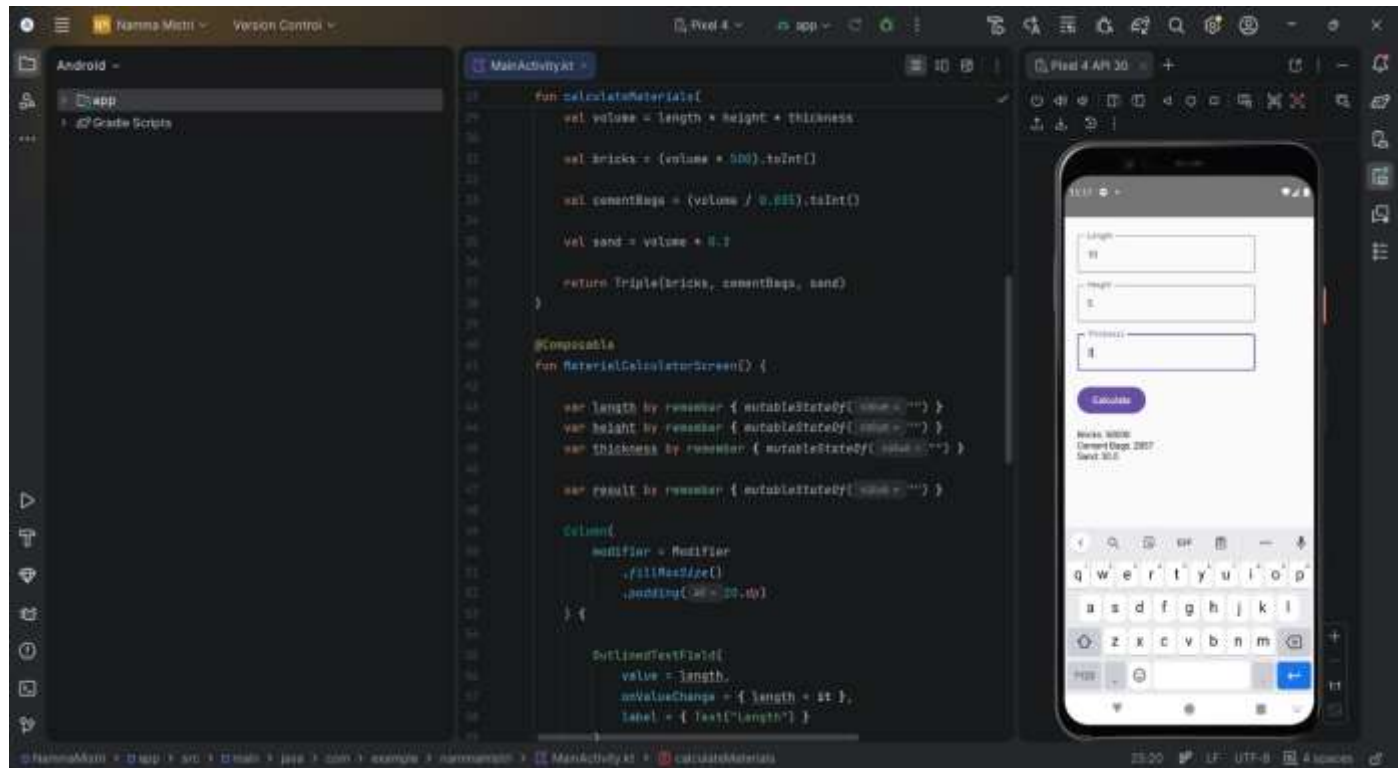


Figure 5.5.1: Android Studio Implementation of Namma-Mistri App

Core Kotlin Implementation for Material Estimation:

```
package com.example.nammamistri

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.compose.foundation.layout.*
import androidx.compose.material3.*
import androidx.compose.runtime.*
import androidx.compose.ui.Modifier
import androidx.compose.ui.unit.dp
class MainActivity : ComponentActivity() {
```

```

override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)
    setContentView {
        MaterialCalculatorScreen()
    }
}

fun calculateMaterials(
    length: Double,
    height: Double,
    thickness: Double
): Triple<Int, Int, Double> {
    val volume = length * height * thickness
    val bricks = (volume * 500).toInt()
    val cementBags = (volume / 0.035).toInt()
    val sand = volume * 0.3
    return Triple(bricks, cementBags, sand)
}

@Composable
fun MaterialCalculatorScreen() {
    var length by remember { mutableStateOf("") }
    var height by remember { mutableStateOf("") }
    var thickness by remember { mutableStateOf("") }
    var result by remember { mutableStateOf("") }

    Column( modifier = Modifier
        .fillMaxSize()
        .padding(20.dp) ) {
        OutlinedTextField(
            value = length,
            onValueChange = { length = it },
            label = { Text("Length") } )
        Spacer(modifier = Modifier.height(10.dp))
        OutlinedTextField(

```



```

        value = height,
        onChange = { height = it },
        label = { Text("Height") }
    )
    Spacer(modifier = Modifier.height(10.dp))
    OutlinedTextField(
        value = thickness,
        onChange = { thickness = it },
        label = { Text("Thickness") }
    )
    Spacer(modifier = Modifier.height(20.dp))
    Button(onClick = {
        val data = calculateMaterials(
            length.toDouble(),
            height.toDouble(),
            thickness.toDouble()
        )
        result = "Bricks: ${data.first}\nCement Bags: ${data.second}\nSand: ${data.third}"
    }) {
        Text("Calculate")
    }
    Spacer(modifier = Modifier.height(20.dp))
    Text(result)
}
}

```

5.6 Scope of the Project

The scope of the Namma-Mistri project mainly includes construction material estimation, labor management, and construction progress monitoring. The application can be used by local builders, small contractors, and rural mistris for handling day-to-day construction activities more efficiently.

The material calculator module allows users to calculate required quantities of bricks, cement, and sand for walls and rooms based on dimensions entered by the user. The labor diary module helps maintain worker attendance and payment records digitally. The site photo feature helps users maintain visual documentation of work progress. The standard rates module allows users to store and update local construction material prices for budgeting and cost planning purposes.

The project is mainly suitable for small-scale and medium-scale construction works such as residential buildings, house construction, and renovation projects. The application is intended to provide a low-cost and accessible digital solution for practical construction-site management.

5.7 Advantages

The Namma-Mistri application provides several advantages for local builders and rural construction workers. One of the major advantages of the system is accurate material estimation. The application helps mistris calculate the required quantity of bricks, cement, and sand based on proper dimensions, which reduces material wastage and unnecessary expenses. This improves overall construction planning and helps complete projects more efficiently.

Another important advantage of the system is digital labor management. The application allows builders to maintain worker attendance, wage records, advance payments, and balance calculations in an organized manner. This reduces confusion in labor payment management and improves financial transparency at construction sites.

The system also saves time and reduces manual effort because calculations and records are maintained automatically within the application. The site photo management feature helps users monitor construction progress and maintain visual proof of work completion. Additionally, the application is simple, user-friendly, and suitable for local builders who may not have advanced technical knowledge.

Overall, the system improves construction-site organization, reduces human error, and provides a practical low-cost digital solution for small-scale construction management.

5.8 Disadvantages

Although the Namma-Mistri application provides many useful features, it also has certain limitations. The application mainly depends on user-entered values, so incorrect input dimensions may lead to inaccurate material estimation results. The system also requires a smartphone and basic knowledge of mobile applications, which may be difficult for some users in rural areas who are not familiar with technology.

Another limitation is that the application is mainly designed for small-scale and medium-scale construction projects. It may not support highly complex calculations required for large commercial construction projects.

The application also depends on proper device storage for maintaining photos and records, and data may be lost if the mobile device is damaged or reset without backup support.

In addition, internet-based features or cloud synchronization may require network connectivity, which can sometimes be limited in rural areas. Therefore, the application may need future improvements for advanced functionality and better scalability.

5.9 Applications

The Namma-Mistri application can be used in various construction-related environments where simple and organized management is required. The system is mainly useful for local builders, rural mistris, and small contractors who handle house construction and small civil projects.

The application can be used for residential building construction where accurate estimation of construction materials is necessary. It can also be used during renovation and repair works to calculate material requirements and maintain labor records properly. Small-scale contractors can use the system for tracking worker attendance and managing daily wages efficiently.

The site photo management feature makes the application useful for monitoring construction progress and maintaining transparency between builders and house owners. Educational institutions and civil engineering students can also use the application for understanding practical construction estimation and management concepts.

Overall, the application is suitable for real-world construction-site management where low-cost, mobile-based digital assistance is required.

5.10 Conclusion

Namma-Mistri is a practical Android-based construction assistant application developed to simplify construction-site management for local builders and rural mistris. The project mainly focuses on solving real-world problems related to material estimation, labor management, and construction progress tracking.

The system helps users calculate construction materials accurately, maintain digital labor records, and organize site activities more efficiently. By reducing manual calculations and improving record management, the application helps save time, reduce wastage, and improve overall construction planning.

The project successfully combines Android technology with practical construction management concepts to create a useful and user-friendly solution for small-scale construction environments. The application demonstrates real-world problem solving, mobile application development, and efficient construction-site assistance through digital technology.

CHAPTER 6

CONCLUSION / SUMMARY

CHAPTER-6

CONCLUSION / SUMMARY

6.1 Overall Internship Summary

The internship conducted at Mind Matrix provided valuable practical exposure to Android application development, AI-assisted workflows, and project implementation methodologies. The internship followed a structured learning process that gradually improved technical knowledge and practical development ability.

6.2 Technical Growth

The internship improved practical understanding of:

Kotlin programming

Android Studio workflows

UI design using Jetpack Compose

GitHub repository management

Debugging methodologies

Application testing processes

6.3 Professional Development

The internship also improved professional skills such as:

Technical communication

Project documentation

Presentation ability

Problem-solving skills

Time management

Exposure to Agile methodology and project management practices improved understanding of real-world software development workflows.

6.4 Final Conclusion

Overall, the internship was highly beneficial in improving technical proficiency, development confidence, practical implementation ability, and understanding of modern software development practices. The knowledge and experience gained during the internship will be valuable for future academic projects, placements, and professional software development careers.